

Inverse Reality Ontology: A Chrono-Causal Framework for Reconstructing Construction Histories from Terminal Physical States

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Abstract

This working paper proposes an inverse, evidence-constrained framework for reconstructing the formation history of a physical object from a terminal state. The Great Pyramid of Khufu is used as the primary archaeological laboratory because it combines large-scale geometry, heterogeneous materials, internal cavities, quarry provenance, structural constraints, muographic observations and unresolved construction chronology. Rather than generating a single narrative of construction, the framework seeks a probability distribution over physically and archaeologically admissible predecessor states and typed partial-order construction graphs. A central contribution is the explicit dual representation of positive material structure and negative architectural space. Architectural voids are represented both topologically and through a counterfactual absent-mass field relative to a declared solid reference; this quantity is not gravitational negative mass. The framework also introduces inverse gravitational bookkeeping, in which gravity retains its physical direction while gravitational work and potential-energy increments are accounted for along an inferred reverse trajectory. Vaidya collapse and equatorial Kerr null-geodesic accessibility are used as mathematical comparison systems to investigate complement-defined regions, critical boundaries and changes of accessibility. No physical equivalence between monumental architecture and black-hole physics is claimed. The intended outcome is a falsifiable computational archaeology platform in which inferred histories generate observable predictions that can be tested against geometry, structural mechanics, geology, archaeology and muography.

Keywords: Egyptology; Egyptian archaeology; Great Pyramid of Giza; Khufu; pyramid construction; inverse archaeology; computational archaeology; archaeological inference; 4D reconstruction; partial-order reconstruction; negative space; topology; persistent homology; muography; ScanPyramids; inverse problems; structural mechanics; causal graphs; Vaidya spacetime; Kerr geodesics; PRAXIOS; Meta-Harness.

1. Research problem

Let X_T denote an as-built terminal object and Y the set of observations available to the researcher. Classical reconstruction commonly begins from candidate narratives and asks whether they are plausible. Here we invert the direction: given X_T , Y , and physical/archaeological constraints L , infer the set of predecessor-state trajectories and construction partial orders that could have generated the object.

The target is therefore not $\text{history} = f(\text{object})$ but a posterior distribution $P(\text{history} \mid \text{object}, \text{evidence}, \text{laws})$. Non-identifiability is treated as a property of the problem rather than an invitation to choose a compelling narrative.

2. Terminal state and inverse coordinate

Define inverse time $\tau = T - t$, with $\tau = 0$ at completion. The inverse engine does not assert that nature literally evolves backward. It computes predecessor states that, under forward physical laws, are compatible with the current state. First derivatives change sign under this coordinate transformation; gravity does not.

3. Material and negative architectural objects

Let $M(t)$ be the occupied material domain and $V(t) = D \setminus M(t)$ its complement in a declared construction domain D . A chamber is not adequately described as empty space. It has a genealogy: reservation, boundary emergence, closure, structural completion and encapsulation. We represent material and voids as coupled complexes K^+ and K^- .

For an architectural cavity V_C , define a counterfactual absent mass $m_C^{(absent)} = -\int_{V_C} \rho_{ref}(x) dV$, where ρ_{ref} is the density of a declared solid reference. This can interface naturally with density-sensitive methods such as muography while avoiding the incorrect claim of physical negative mass.

4. Inverse gravitational bookkeeping

For a component transported from source elevation z_s to final elevation z_f , its forward change in gravitational potential is $\Delta U = m g(z_f - z_s)$. On the inferred inverse path the signed bookkeeping term is $\Delta U^{\text{inv}} = -\Delta U$. This is a reverse-accounting variable, not antigravity. For a geometrically ideal uniform pyramid, analytic mass and potential-energy integrals provide useful baseline constraints before block-level structural modeling.

5. Typed causal graph

The object induces a typed dependency graph. Edges can encode structural support, geometry, access, load transfer, void preservation, provenance, and procedural precedence. An element is inverse-removable only when downstream dependencies have been removed and the remaining state satisfies stability, accessibility and evidence constraints. Consequently, elevation is not chronology: the desired unknown is a 4D construction-time field $T(x, y, z)$ plus partial-order uncertainty.

6. Khufu as laboratory

Published muography demonstrates that internal density anomalies can reveal previously unknown voids, including the ScanPyramids Big Void and the north-face corridor. AERA quarry observations constrain local stone supply and volumes. Recent paleohydrological work on the Ahramat Branch constrains the broader transport environment. These independent evidence domains make Khufu particularly suitable for an inverse multi-constraint reconstruction.

The King's Chamber is an instructive first subgraph. Its physical floor, granite walls, ceiling beams, relieving system, gabled superstructure, access route and surrounding masonry should be represented as dependent entities. Under inverse disassembly, the chamber retains topological identity until removal of its enclosing boundary reconnects the void to the exterior. Thus geometrical emptiness, topological identity and architectural identity are distinct state variables.

7. Complement-defined objects and accessibility

We define a general complement object $N_R(\lambda) = \Omega \setminus \text{Accessible}_R(\Omega, \lambda)$, where relation R specifies the admissible connection. This abstraction can describe a material cavity, a causal non-escape region, or a forbidden subset of geodesic phase space. It is a mathematical comparison, not an assertion that these systems share the same physical mechanism.

8. Relativistic comparison systems

In a didactic Vaidya collapse, the apparent horizon responds locally to the mass function while the event horizon is global and can be reconstructed by backward integration after imposing a future condition. This offers a useful analogy for terminal-state constraints on antecedent accessibility.

For equatorial null geodesics in Kerr, the radial potential $R(r; b)$ defines allowed radial regions through $R \geq 0$. Critical light rings satisfy $R=0$ and $dR/dr=0$, forming separatrices at which the connectedness of the allowed radial set can change. This motivates a generic critical-boundary template $\Phi=0$, $\text{grad } \Phi=0$ for investigating topology-changing events in an accessibility space.

9. Falsifiability and Meta-Harness

The inverse engine must generate predicted evidence. If candidate history H predicts geometry, marks, density patterns, quarry signatures or structural traces $\hat{Y}(H)$, the framework compares them against observed evidence Y . Meta-Harness should preserve competing histories, provenance, contradictions and uncertainty instead of collapsing the posterior into a single authoritative narrative.

10. Research program

Priority work includes: high-resolution as-built geometry; block and contact graph extraction; explicit King's Chamber dependency DAG; negative-space persistent topology; finite-element load paths; lithological/provenance distributions; quarry-to-monument material balance; access/routing constraints; Bayesian partial-order inference; and muographic forward models for candidate void/construction hypotheses.

11. Scope and limitations

This paper presents a research architecture. It does not solve the historical construction method of Khufu, does not claim new gravitational physics, and does not treat black-hole analogies as archaeological evidence. Its scientific value depends on whether the framework produces discriminating, testable predictions that reduce the set of admissible construction histories.

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